

JY Liu

Solutions Architect | Technical Lead | Senior Full-Stack Engineer

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Summary

- Solutions architect and technical lead with 12+ years shipping web, mobile, cloud, and integration-heavy platforms across fintech, SaaS, banking, consular, and enterprise environments.
- Hands-on across frontend, backend, APIs, DevOps, and cloud infrastructure — consistently taking products from discovery and architecture through production delivery.
- Leverages AI tooling pragmatically across the SDLC — analysis, implementation, prototyping, and quality assurance — while building AI-native products end-to-end and maintaining production-grade technical rigor.

Experience

- Vailley AI Solution Pte. Ltd.**, Senior Web Developer (Technical Lead) Remote (Singapore)
Jun 2025 – Present
- Leading cross-functional delivery for VFS Global's visa and consular platforms — combining solution architecture with hands-on implementation across the application stack, integration layer, and platform.
 - Delivered a Consular Dashboard with RBAC and ABAC controls, an ADR portal with structured document submission, ETL-backed operational visibility, a digital corporate visa platform for agents, and SAP Concur to OneVasco integration for VFS Global.
- Shift Markets**, Senior Software Engineer Remote (NY, US)
Jan 2023 – Present
- Lead engineer for CXM Direct (worldwide multi-asset trading platform) and CRM initiatives, owning backend delivery with ThinkPHP and frontend execution in TypeScript.
 - Engineered a Unified Payment Gateway integrating 12 PSPs across 12+ regions and 17+ currencies, supporting 235,000+ registered users across CXMD (Singapore) and CXMT (Tokyo) trading platforms.
 - Strengthened system reliability and delivery velocity through architectural cleanup, perf tuning, and production-oriented engineering practices.
- InternetBrands**, Senior Web Engineer Remote (California, US)
Apr 2023 – Sept 2024
- Maintained and improved the Dentrux Hub Forms (Online Form for Dentrux & HOP) Laravel backend — patient form management platform for dental practices, focused on reliability, performance, and scale.
 - Introduced PHP linting and unit testing standards, growing test coverage from 0% to 90% across the codebase.
 - Managed Kubernetes deployment with Helm and Rancher for the backend.
- MaiDaLv**, Senior Full Stack Developer Remote (Guangzhou, China)
Feb 2025 – Apr 2025
- Built Spring Boot, UniApp, Vue.js, and Python-based product workflows covering subs, WeChat integration, and IP-risk scan.
- Cake Experiential Communications Asia**, Solutions Architect Remote (KL, Malaysia)
Apr 2021 – May 2023
- Led end-to-end architecture and delivery for Vue.js and Laravel platforms used in regional enterprise and campaign engagements.
 - Introduced engineering standards — TDD, API governance, and backend scaling practices — for client-critical systems.
- Glints**, Web Software Engineer Remote (Singapore)
Jan 2022 – Dec 2022
- Built full-stack recruiter tooling for Superpowered using Next.js and Koa.js, with emphasis on workflow efficiency and usable internal product design.
 - Strengthened dashboard responsiveness and release quality through improved testing and deployment automation.

InnoLab Global , Senior Information Technology Consultant - Delivered cross-platform products across Angular, Ionic, Flutter, Firebase, and AWS while automating cloud delivery workflows and mentoring junior developers across multiple client engagements. - Shipped the Forest City Super App on a Turborepo + Next.js + Prisma + PostgreSQL + Auth.js stack covering community, events, payments, partner portal, and admin console features for residents and partners. - Built Aegis Shooting Academy (ASA) digital platform end-to-end with NestJS + Prisma backend, Next.js admin dashboard, and React Native/Expo mobile application. - Delivered a corporate website and WooCommerce product catalogue for 3V Malaysia (furniture manufacturer) using Docker Compose with a premium Apple-inspired theme. - Architected and led delivery for AvaEra's compensation platform — modernized legacy stack into pnpm + Turborepo monorepo with Next.js, Prisma, PostgreSQL, and a 15-service reward engine covering direct sponsor, team bonus, pool sharing, profit distribution, and merchant commissions. - Built and shipped HappyAI for Gembira Bina Sdn Bhd, an AI-powered tender document extraction and validation platform — Next.js + FastAPI + SQLite with Groq LLM primary / Ollama fallback, confidence-badged review UI, and deterministic batch ingest pipeline.	Remote (KL, Malaysia) Sept 2018 – Present
Supahands , Full Stack Developer Built full-stack features for Supahands' distributed-team product.	Remote (KL, Malaysia) Feb 2018 – Jun 2018
The Consociation LLP , Information Technology Consultant Led full-stack delivery for products including iHeartCity and Swiiipe using Angular, NestJS, GraphQL, and Firebase with AWS-based DevOps automation.	Remote (Singapore) Sept 2016 – Apr 2020
Buddy Holdings Pte Ltd , Senior Information Technology Consultant - Architected and led delivery of the Buddy System Asia MLM platform serving 2,000+ active members across Singapore, Malaysia, China, and Indonesia — Angular + Laravel + MySQL + Kafka covering hierarchy management, point calculations, point redemption, inventory tracking, and family tree updates at scale. - Built agent-facing UI for real-time tracking of progress, points, and family trees across the multi-country user base.	Remote (Singapore) Sept 2016 – Jan 2024
Fuji Xerox Pte Ltd , Embedded Software Engineer Developed embedded software for printer/scanner hardware platforms with C/C++.	Singapore Feb 2016 – May 2016
Netccentric Pte Ltd , Software Developer Built full-stack web features for digital ad campaigns.	Singapore Oct 2015 – Dec 2015
Nexstream Sdn Bhd , Hybrid App Developer Built hybrid mobile apps for iOS and Android using web technologies.	Kuala Lumpur, Malaysia Jun 2014 – Jul 2015

Selected Projects

Unified Payment Gateway Specialized payment infrastructure unifying 20+ global Payment Service Providers (PSPs) into a single, high-concurrency API layer. - Architected and delivered a unified integration layer for 12 PSPs across 12+ regions and 17+ currencies, handling complex transaction workflows, reconciliation, and automated settlement for 235,000+ platform users. - Secured financial operations through strict audit logging, sensitive data isolation, and robust error-recovery mechanisms.	Jan 2023 – Present
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VFS Global Consular Platform

Jun 2025 – Present

Mission-critical visa and consular services platform delivering digital transformation for global diplomatic missions.

- Delivered a high-security Consular Dashboard with granular RBAC/ABAC and integrated SAP Concur for automated expense and travel management, supporting multi-country diplomatic operations.
- Built a digital corporate visa platform enabling agents to manage large-scale applications with real-time tracking and ETL-backed analytics.

Superpowered

Jan 2022 – Dec 2022

Talent platform delivered with scalable web architecture, backend APIs, GraphQL, PostgreSQL, and event-driven workflows.

- Built a responsive Next.js and Koa.js application with GraphQL integration, backend optimization, and scalable recruiter tools.

Dentrix Hub Forms (Online Form for Dentrix & HOP)

Apr 2023 – Sept 2024

Dental practice patient form management platform — Laravel backend with Kubernetes deployment, PHP linting, and unit testing standards.

- Maintained and improved the Laravel backend, strengthening response times and concurrency handling for patient form workflows.
- Introduced PHP linting and unit testing, growing test coverage from 0% to 90%.
- Managed Kubernetes deployment with Helm and Rancher for scalable, reliable ops.

Education

Master of Business Administration University of Malaya, Business Analytics
(Expected Oct 2026)

Kuala Lumpur, Malaysia
Oct 2023 – Oct 2026

Master of Computer Science University of Malaya, Mobile Cloud Computing

Kuala Lumpur, Malaysia
Jan 2011 – Jan 2013

Bachelor of Computer Science Universiti Sains Malaysia, Software Engineering

Malaysia
Jan 2008 – Jan 2011

Skills

Full Stack Engineering: TypeScript, JavaScript, PHP, Python, Java, Laravel, ThinkPHP, React, Next.js, Angular, Vue.js, NestJS, Node.js, Spring Boot, Flutter, Ionic

Architecture, Cloud & DevOps: Solution architecture, AWS, GCP, Firebase, Docker, Kubernetes, Helm, Rancher, CI/CD, deployment automation, integration-heavy platform delivery

Backend, Data & Integration: PostgreSQL, MySQL, Redis, Apache Kafka, GraphQL, REST APIs, microservices, ETL workflows, event-driven and serverless systems

AI-Enabled Delivery & Multi-Agent Systems: Orchestrating LLM-based agents, RAG pipelines, prompt engineering, and custom GPT development to accelerate product delivery.

Languages

English, Malay, Cantonese: Professional

Mandarin: Native